

31.2 When ordinary least squares is best

The analysis in Example (A trend in sea surface temperature) left a puzzle: the generalized and the ordinary estimates of the trend agreed to three decimals and the seasonal coefficients to four, yet their standard errors disagreed by a factor of more than two in one direction and by half as much again in the other. Both halves have one explanation, the subject of this section: for a large class of designs ordinary least squares is *already* the best linear unbiased estimator, or very nearly so, while its reported standard error is wrong all the same.

Throughout, $\mathbf{V}$ is nonnegative definite and may be singular, $\mathbf{M}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, and “ $\mathbf{AY}$ is a BLUE of $\mathbf{X}\boldsymbol{\beta}$ ” means that $\boldsymbol{\rho}^\top \mathbf{AY}$ is best linear unbiased for $\boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$ at every $\boldsymbol{\rho} \in \mathbb{R}^n$.

31.2.1 A criterion for a best estimator

Everything in this section, and most of the next, rests on one small lemma, the covariance version of Exercise (C3): a linear estimator is best exactly when it is uncorrelated with every linear statistic that estimates zero. What is new is that $\mathbf{V}$ may be singular, that the criterion is read off one vector $\mathbf{c}$ at a time, and that uniqueness holds on $\mathcal{C}(\mathbf{X} : \mathbf{V})$, not merely with probability one.

Lemma 31.2.1. Criterion for a BLUE

Assume Eq. 31.1.1 with $\mathbf{V}$ nonnegative definite. Let $\mathbf{c} \in \mathbb{R}^n$ and $\boldsymbol{\lambda} = \mathbf{X}^\top \mathbf{c}$, so that $\mathbf{c}^\top \mathbf{Y}$ is a linear unbiased estimator of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$.

- (a) $\mathbf{c}^\top \mathbf{Y}$ is a BLUE of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ if and only if $\mathbf{V}\mathbf{c} \in \mathcal{C}(\mathbf{X})$.
- (b) Any two BLUEs of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ are equal with probability one. More precisely, if $\mathbf{c}^\top \mathbf{Y}$ and $\mathbf{b}^\top \mathbf{Y}$ are both BLUEs then $\mathbf{c}^\top \mathbf{y} = \mathbf{b}^\top \mathbf{y}$ for every $\mathbf{y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})$.
- (c) A matrix estimator $\mathbf{AY}$ with $\mathbf{AX} = \mathbf{X}$ is a BLUE of $\mathbf{X}\boldsymbol{\beta}$ if and only if $\mathbf{AV}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$, equivalently $\mathcal{C}(\mathbf{VA}^\top) \subseteq \mathcal{C}(\mathbf{X})$.

Proof.

- (a) Every linear unbiased estimator of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is $(\mathbf{c} + \mathbf{d})^\top \mathbf{Y}$ with $\mathbf{X}^\top \mathbf{d} = \mathbf{0}$, that is, with $\mathbf{d} \in \mathcal{C}(\mathbf{X})^\perp$. For real t ,

$$\text{Var}((\mathbf{c} + t\mathbf{d})^\top \mathbf{Y}) = \sigma^2(\mathbf{c}^\top \mathbf{V}\mathbf{c} + 2t\mathbf{c}^\top \mathbf{V}\mathbf{d} + t^2\mathbf{d}^\top \mathbf{V}\mathbf{d}),$$

a quadratic in t with nonnegative leading coefficient. If $\mathbf{c}^\top \mathbf{Y}$ is best, this quadratic is minimized at $t = 0$ for every such $\mathbf{d}$, which forces $\mathbf{c}^\top \mathbf{V}\mathbf{d} = 0$. Conversely, if $\mathbf{c}^\top \mathbf{V}\mathbf{d} = 0$ for every $\mathbf{d} \perp \mathcal{C}(\mathbf{X})$, then $\text{Var}((\mathbf{c} + \mathbf{d})^\top \mathbf{Y}) = \text{Var}(\mathbf{c}^\top \mathbf{Y}) + \sigma^2\mathbf{d}^\top \mathbf{V}\mathbf{d} \geq \text{Var}(\mathbf{c}^\top \mathbf{Y})$. So $\mathbf{c}^\top \mathbf{Y}$ is best iff $\mathbf{V}\mathbf{c} \perp \mathcal{C}(\mathbf{X})^\perp$, which is $\mathbf{V}\mathbf{c} \in \mathcal{C}(\mathbf{X})$.

- (b) Let $\mathbf{c}^\top \mathbf{Y}$ and $\mathbf{b}^\top \mathbf{Y}$ be BLUEs, with common variance v and covariance κ . The average $\frac{1}{2}(\mathbf{c} + \mathbf{b})^\top \mathbf{Y}$ is unbiased, so $v \leq \frac{1}{4}(v + v + 2\kappa)$, that is $v \leq \kappa$. Therefore

$$\text{Var}((\mathbf{c} - \mathbf{b})^\top \mathbf{Y}) = 2v - 2\kappa \leq 0,$$

so it is zero, and $(\mathbf{c} - \mathbf{b})^\top \mathbf{Y}$ has mean zero and zero variance: it vanishes with probability one. For the deterministic statement, zero variance means $(\mathbf{c} - \mathbf{b})^\top \mathbf{V}(\mathbf{c} - \mathbf{b}) = 0$,

hence $\mathbf{V}(\mathbf{c} - \mathbf{b}) = \mathbf{0}$ because $\mathbf{V}$ is nonnegative definite, so $\mathbf{c}$ and $\mathbf{b}$ agree on $\mathcal{C}(\mathbf{V})$; and $\mathbf{X}^\top(\mathbf{c} - \mathbf{b}) = \mathbf{0}$ makes them agree on $\mathcal{C}(\mathbf{X})$. They therefore agree on $\mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{X} : \mathbf{V})$.

- (c) Apply (a) to $\mathbf{c} = \mathbf{A}^\top \boldsymbol{\rho}$, which satisfies $\mathbf{X}^\top \mathbf{A}^\top \boldsymbol{\rho} = (\mathbf{A}\mathbf{X})^\top \boldsymbol{\rho} = \mathbf{X}^\top \boldsymbol{\rho}$, so $\mathbf{c}^\top \mathbf{Y}$ is unbiased for $\boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$. The condition $\mathbf{V}\mathbf{A}^\top \boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})$ for every $\boldsymbol{\rho}$ is $\mathcal{C}(\mathbf{V}\mathbf{A}^\top) \subseteq \mathcal{C}(\mathbf{X})$, which says $(\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{A}^\top = \mathbf{0}$; transposing gives $\mathbf{A}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$.

■

31.2.2 The characterization

Theorem 31.2.2. When ordinary least squares is best

Assume Eq. 31.1.1 with $\mathbf{V}$ nonnegative definite, and let $\mathbf{W}$ be any $n \times (n - r)$ matrix with $\mathcal{C}(\mathbf{W}) = \mathcal{C}(\mathbf{X})^\perp$.

- (a) The following are equivalent.
- (i) $\mathbf{M}\mathbf{Y}$ is a BLUE of $\mathbf{X}\boldsymbol{\beta}$.
 - (ii) $\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$, equivalently $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{M}$.
 - (iii) $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$.
 - (iv) $\mathbf{V} = \mathbf{X}\mathbf{A}\mathbf{X}^\top + \mathbf{W}\mathbf{B}\mathbf{W}^\top$ for some nonnegative definite $\mathbf{A}$ and $\mathbf{B}$.
 - (v) $\mathcal{C}(\mathbf{X})$ has an orthonormal basis of eigenvectors of $\mathbf{V}$.
- (b) For $\boldsymbol{\rho} \in \mathbb{R}^n$, the estimator $\boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$ of $\boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$ is a BLUE if and only if $\mathbf{V}\mathbf{M}\boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})$. When $\text{rank}(\mathbf{X}) = p$ and $\boldsymbol{\lambda} \in \mathbb{R}^p$, this reads $\mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X})$.
- (c) If $\mathbf{V}$ is positive definite, the conditions of (a) are also equivalent to $\mathbf{M}\mathbf{y} = \mathbf{A}\mathbf{v}\mathbf{y}$ for every $\mathbf{y}$, and then ordinary and generalized least squares give the same fit, the same estimate of every estimable function, and the same $\text{SSE}_{\mathbf{V}}$.

Proof. For positive definite $\mathbf{V}$ the chain (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) is Exercise (C3); the argument below needs no invertibility.

- (a) (i) $\Leftrightarrow$ (ii). Take $\mathbf{A} = \mathbf{M}$ in Lemma 31.2.1(c), which applies because $\mathbf{M}\mathbf{X} = \mathbf{X}$. For the second form of (ii): if $\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ then $\mathbf{M}\mathbf{V} = \mathbf{M}\mathbf{V}\mathbf{M}$, and $\mathbf{M}\mathbf{V}\mathbf{M}$ is symmetric, so $\mathbf{M}\mathbf{V}$ is symmetric and equals $(\mathbf{M}\mathbf{V})^\top = \mathbf{V}\mathbf{M}$. Conversely $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{M}$ gives $\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{V}\mathbf{M} - \mathbf{V}\mathbf{M} = \mathbf{0}$.
- (ii) $\Leftrightarrow$ (iii). If $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{M}$ then $\mathbf{V}\mathbf{X} = \mathbf{V}\mathbf{M}\mathbf{X} = \mathbf{M}\mathbf{V}\mathbf{X} \in \mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$. Conversely, if $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ then $\mathbf{M}\mathbf{V}\mathbf{X} = \mathbf{V}\mathbf{X}$. Since $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$, every column of $\mathbf{M}$ is a combination of columns of $\mathbf{X}$, so $\mathbf{M}\mathbf{V}\mathbf{M} = \mathbf{V}\mathbf{M}$. The left side is symmetric, so $\mathbf{V}\mathbf{M} = (\mathbf{V}\mathbf{M})^\top = \mathbf{M}\mathbf{V}$.
- (iii) $\Rightarrow$ (iv). Split the identity as $\mathbf{I} = \mathbf{M} + (\mathbf{I} - \mathbf{M})$:

$$\begin{aligned} \mathbf{V} &= \mathbf{M}\mathbf{V}\mathbf{M} + \mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) \\ &\quad + (\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{M} + (\mathbf{I} - \mathbf{M})\mathbf{V}(\mathbf{I} - \mathbf{M}) \\ &= \mathbf{M}\mathbf{V}\mathbf{M} + (\mathbf{I} - \mathbf{M})\mathbf{V}(\mathbf{I} - \mathbf{M}), \end{aligned}$$

the cross terms vanishing by (ii). Now $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ by Theorem 6.3.6, so $\mathbf{M}\mathbf{V}\mathbf{M} = \mathbf{X}\mathbf{A}\mathbf{X}^\top$ with $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}$, which is nonnegative definite

because $\mathbf{V}$ is. Likewise $\mathbf{I} - \mathbf{M} = \mathbf{W}(\mathbf{W}^\top \mathbf{W})^{-1} \mathbf{W}^\top$, since $\mathbf{W}$ has full column rank and spans $\mathcal{C}(\mathbf{X})^\perp$, so $(\mathbf{I} - \mathbf{M})\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{W}\mathbf{B}\mathbf{W}^\top$ with $\mathbf{B} = (\mathbf{W}^\top \mathbf{W})^{-1} \mathbf{W}^\top \mathbf{V}\mathbf{W}(\mathbf{W}^\top \mathbf{W})^{-1}$, also nonnegative definite.

- (iv) $\Rightarrow$ (iii). $\mathbf{W}^\top \mathbf{X} = \mathbf{0}$, so $\mathbf{V}\mathbf{X} = \mathbf{X}\mathbf{A}\mathbf{X}^\top \mathbf{X} \in \mathcal{C}(\mathbf{X})$.
- (v) $\Leftrightarrow$ (v). Condition (iii) says $\mathcal{C}(\mathbf{X})$ is invariant under $\mathbf{V}$. If it is, the restriction of $\mathbf{V}$ to $\mathcal{C}(\mathbf{X})$ is a symmetric operator on that subspace, so by Theorem 1.7.2 it has an orthonormal eigenbasis there, and these are eigenvectors of $\mathbf{V}$. Conversely, if $\mathcal{C}(\mathbf{X})$ is spanned by eigenvectors of $\mathbf{V}$ then $\mathbf{V}$ maps each of them, and hence $\mathcal{C}(\mathbf{X})$, into $\mathcal{C}(\mathbf{X})$.
- (b) By Lemma 31.2.1(a) with $\mathbf{c} = \mathbf{M}\boldsymbol{\rho}$, which satisfies $\mathbf{X}^\top \mathbf{M}\boldsymbol{\rho} = \mathbf{X}^\top \boldsymbol{\rho}$, so that $\boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$ is unbiased for $\boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$. For the full-rank form, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$ with $\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, and then $\mathbf{M}\boldsymbol{\rho} = \boldsymbol{\rho}$.
- (c) With $\mathbf{V}$ positive definite, (iii) is Kruskal's condition, and Theorem 6.9.3 says it holds iff $\mathbf{M}\mathbf{y} = \mathbf{A}_\mathbf{V}\mathbf{y}$ for every $\mathbf{y}$. Equal fits give equal estimates of every estimable function by Theorem 6.4.4, and equal residuals, hence equal $\text{SSE}_\mathbf{V}$.

■

Form (iv) is Rao's **simple covariance structure**: the error splits as $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_1 + \boldsymbol{\varepsilon}_2$ with $\boldsymbol{\varepsilon}_1 \in \mathcal{C}(\mathbf{X})$, $\boldsymbol{\varepsilon}_2 \perp \mathcal{C}(\mathbf{X})$ and the two parts uncorrelated (Exercise (B1)). Ordinary least squares is best exactly when the noise splits along the model space in this way, for then the residuals carry no usable information about $\boldsymbol{\beta}$. Form (v) is the one to check when the eigenstructure of $\mathbf{V}$ is known.

31.2.3 Balanced layouts and a common shock

The most important case in practice is a balanced layout whose errors share a shock at some level of the design.

Example. Equicorrelation in a balanced layout

Let the $n = Gm$ observations fall into G groups of equal size m , let $\mathbf{Z}$ be the $n \times G$ matrix of group indicators, and let $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$, the covariance produced by a group-level shock. Since $\mathbf{Z}\mathbf{Z}^\top$ is m times the orthogonal projection onto the group-constant vectors, condition (iii) holds iff $\mathcal{C}(\mathbf{X})$ is closed under replacing a vector by its group means — which happens whenever $\mathcal{C}(\mathbf{X})$ is spanned by group-constant vectors together with vectors whose group means vanish, exactly what a balanced design gives through its between-group and within-group contrasts.

In the El Niño data of Example (A trend in sea surface temperature) the groups are years. If the seasonal part of the model is the full set of eleven month effects, every column of $\mathbf{X}$ splits into a year-level part and a part whose year means vanish, both again in $\mathcal{C}(\mathbf{X})$, so the condition holds *exactly*: the two fits agree to nine decimals, whatever ρ . Their standard errors do not. At $\rho = 0.5$ the root of the expected reported variance $E(s^2)\{(\mathbf{X}^\top \mathbf{X})^{-1}\}_{22}$ is only 0.391 of the true standard error of the trend.

The same argument covers randomized complete block designs with random blocks, and any design whose every regressor is constant within a block or sums to zero in each block. The analysis of variance tables of Part IV therefore survive random block effects as *point estimates*; only their error terms need rethinking, and Chapter 33 develops the split-plot case, where condition (iv) becomes a decomposition into error strata. Incomplete block designs break the condition: between-block comparisons then carry treatment information that a within-block

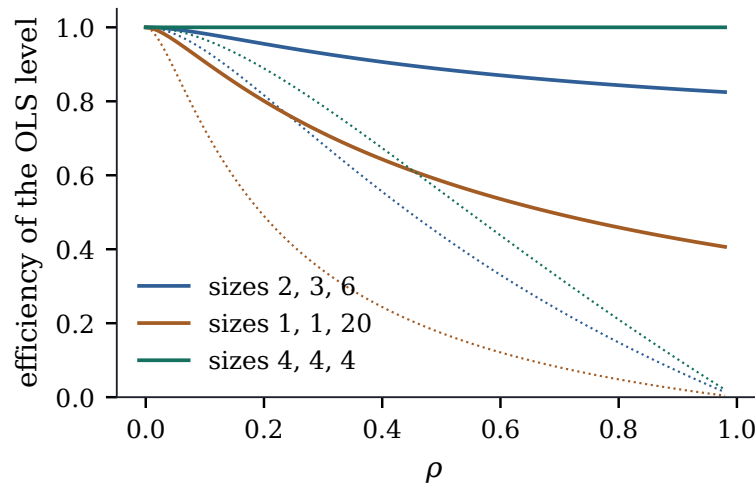


Figure 31.2.1: Efficiency of the ordinary least squares estimate of the overall level, for three cluster layouts with equicorrelation ρ inside clusters (solid). The balanced layout loses nothing at any ρ . The dotted curves are the Kantorovich bound of Theorem 19.2.3 for the same $\mathbf{V}$, far below the efficiency the design actually attains.

analysis discards, and recovering it — Yates’s recovery of interblock information — is generalized least squares.

31.2.4 When some contrasts are best and others are not

Part (b) of Theorem 31.2.2 is finer than part (a): even when $\mathbf{MY}$ is not best for $\mathbf{X}\boldsymbol{\beta}$ as a whole, particular functions still can be. The mechanism is easiest to see when the balance of Example (Equicorrelation in a balanced layout) is broken.

Example. Unequal clusters

Take G clusters of sizes $m_1, \dots, m_G$, a covariate $\mathbf{w}$ centred inside every cluster, and the model $\mathbf{X} = [\mathbf{1}, \mathbf{w}]$ with $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$. Because $\mathbf{1}^\top \mathbf{w} = 0$, the vector attached to the slope is $\mathbf{c} = \mathbf{w}/\mathbf{w}^\top \mathbf{w}$, and $\mathbf{Z}\mathbf{Z}^\top \mathbf{w} = \mathbf{0}$ because $\mathbf{w}$ has zero mean in each cluster; so $\mathbf{V}\mathbf{c} = (1 - \rho)\mathbf{c} \in \mathcal{C}(\mathbf{X})$ and the within-cluster slope is estimated best by ordinary least squares, for every ρ and every set of sizes.

The vector attached to the level is $\mathbf{c} = \mathbf{1}/n$, and $\mathbf{Z}\mathbf{Z}^\top \mathbf{1}$ is the vector whose entry for an observation in cluster k is m_k . That step function lies in $\mathcal{C}(\mathbf{X}) = \text{span}\{\mathbf{1}, \mathbf{w}\}$ only if all the m_k are equal, since $\mathbf{w}$ is not constant within clusters. So with unequal clusters the overall level is *not* estimated best by ordinary least squares; Eq. 31.1.8 gives the estimator that is.

With sizes 2, 3, 6 and $\rho = 0.5$ the weights of Eq. 31.1.8 are 1.333, 1.500 and 1.714, summing to 4.5476, so the generalized estimate of the level has variance $0.2199\sigma^2$ against $0.2479\sigma^2$ for the sample mean: an efficiency of 0.887, well above the Kantorovich bound 0.437 that Proposition 31.2.3(c) allows for the eigenvalues 3.500 and 0.500 of this $\mathbf{V}$. The slope loses nothing. Meanwhile the usual standard errors are wrong for both: the root of the expected reported variance is 0.563 of the true standard error for the level and 1.314 of it for the slope.

```
import numpy as np
```

```

sizes = [2, 3, 6]
w = np.array([-1.0, 1.0, -1.0, 0.0, 1.0, -3.0, -2.0, -1.0, 1.0, 2.0, 3.0])
n = len(w)
Z = np.zeros((n, len(sizes))) # cluster indicators
Z[np.arange(n), np.repeat(np.arange(len(sizes)), sizes)] = 1.0
X = np.column_stack([np.ones(n), w])
rho = 0.5
V = (1 - rho) * np.eye(n) + rho * Z @ Z.T

Ginv = np.linalg.inv(X.T @ X)
Cov_ols = Ginv @ X.T @ V @ X @ Ginv # sigma^2 = 1 throughout
Cov_gls = np.linalg.inv(X.T @ np.linalg.solve(V, X))
print("efficiency of OLS: level %.4f slope %.4f"
      % (Cov_gls[0, 0] / Cov_ols[0, 0], Cov_gls[1, 1] / Cov_ols[1, 1]))

```

[Run this code online](#)

31.2.5 How much is lost, and what is actually broken

Proposition 31.2.3. The cost of ordinary least squares

Assume Eq. 31.1.1 with $\mathbf{V}$ positive definite and $\text{rank}(\mathbf{X}) = p$. Write $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and let $\lambda_1 \geq \dots \geq \lambda_n > 0$ be the eigenvalues of $\mathbf{V}$. For $\boldsymbol{\lambda} \neq \mathbf{0}$ define the efficiency $e(\boldsymbol{\lambda}) = \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}) / \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$.

- (a) $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}}_{\mathbf{V}})$ is nonnegative definite, so $0 < e(\boldsymbol{\lambda}) \leq 1$ for every $\boldsymbol{\lambda}$.
- (b) $e(\boldsymbol{\lambda}) = 1$ if and only if $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}$ with probability one, which by Theorem 31.2.2(b) happens iff $\mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X})$.
- (c) $e(\boldsymbol{\lambda}) \geq 4\lambda_1\lambda_n/(\lambda_1 + \lambda_n)^2$ for every $\boldsymbol{\lambda}$ and every $\mathbf{X}$, and the bound is attained.

Proof.

- (a) Both $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ and $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}$ are linear unbiased estimators of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, and the latter is best by Corollary 7.2.3, so $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \geq \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}) > 0$. Taking $\boldsymbol{\lambda}$ over $\mathbb{R}^p$ gives $\boldsymbol{\lambda}^\top (\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}}_{\mathbf{V}})) \boldsymbol{\lambda} \geq 0$.
- (b) If $e(\boldsymbol{\lambda}) = 1$ then $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ attains the minimum variance and is therefore a BLUE, so it equals $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}$ with probability one by Lemma 31.2.1(b); the converse is immediate. The condition is part (b) of Theorem 31.2.2.
- (c) This is Theorem 19.2.3, proved in Section 19.2 by a matrix form of Kantorovich's inequality, together with the design given there that attains it. ■

Part (c) is the worst case over all designs and all $\boldsymbol{\lambda}$. A determinant version is also known: if $p \leq n/2$, then

$$\frac{\det \text{Cov}(\hat{\boldsymbol{\beta}}_{\mathbf{V}})}{\det \text{Cov}(\hat{\boldsymbol{\beta}})} \geq \prod_{j=1}^p \frac{4\lambda_j\lambda_{n-j+1}}{(\lambda_j + \lambda_{n-j+1})^2},$$

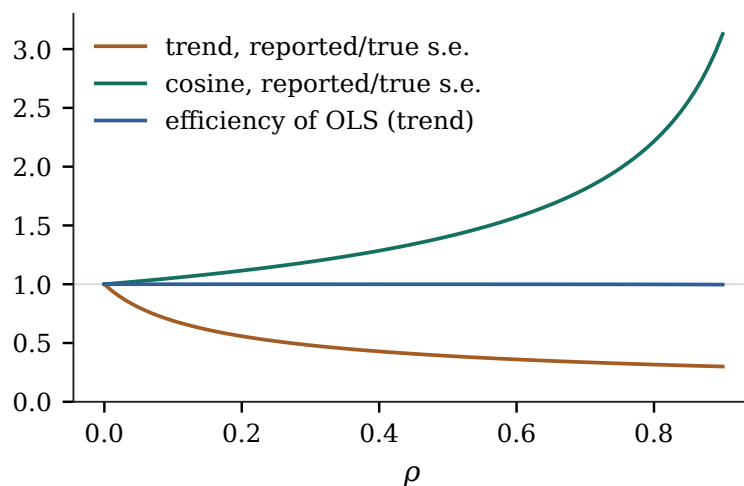


Figure 31.2.2: The El Niño model with three harmonics, as a function of the intraclass correlation ρ . The efficiency of the ordinary least squares trend estimate stays above 0.99, while the ratio of the reported to the true standard error falls to a third for the trend and rises past three for a seasonal coefficient.

with equality for a design built from the corresponding pairs of eigenvectors. This is the Bloomfield–Watson–Knott inequality, proved independently by Bloomfield and Watson (1975) and by Knott (1975) in back-to-back papers. We state it without proof; the argument needs more matrix theory than this book develops.

In practice the loss is far smaller, because real designs are not adversarial. The El Niño model with three harmonics is typical: Kruskal’s condition fails, with $\max_{s,t} | \{ (\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{X} \}_{st} | = 0.126$, yet the efficiency of the trend coefficient is 0.9996 and that of every seasonal coefficient is 1.0000 to four decimals. The intercept is exactly best, because $\mathbf{1}$ is orthogonal to every other column, so $\mathbf{c} = \mathbf{1}/n$ and $\mathbf{V}\mathbf{1} = \{1 + 11\rho\}\mathbf{1}$ is a multiple of $\mathbf{1}$. What is badly wrong is the standard error: the root of the expected value of $s^2\{(\mathbf{X}^\top\mathbf{X})^{-1}\}_{jj}$ is 0.390 of the true standard error for the trend and 1.406 of it for the first cosine, and Figure 31.2.2 shows both ratios running away as ρ grows while the efficiency never falls below 0.99.

Idea. Correlated errors damage the *standard error* far more than the *estimate*. Theorem 19.2.3 bounds the efficiency below by a quantity depending only on the extreme eigenvalues of $\mathbf{V}$, and for ordinary designs the loss is a few per cent. Nothing bounds the error in $s^2(\mathbf{X}^\top\mathbf{X})^{-1}$, which can be too small or too large by any factor. If only one of the two problems can be fixed, fix the standard error.

That is what the sandwich estimator of Section 21.4 does. It leaves $\hat{\boldsymbol{\beta}}$ alone and replaces $s^2(\mathbf{X}^\top\mathbf{X})^{-1}$ by $(\mathbf{X}^\top\mathbf{X})^{-1}(\sum_k \mathbf{X}_k^\top \hat{\boldsymbol{\varepsilon}}_k \hat{\boldsymbol{\varepsilon}}_k^\top \mathbf{X}_k)(\mathbf{X}^\top\mathbf{X})^{-1}$, summed over units that can be treated as independent; for clustered data the unit is a cluster, and the argument of Theorem 21.4.2 applies cluster by cluster rather than observation by observation. We do not prove that extension: Theorem 21.4.2 assumes independent observations, consistency of the cluster form as the number of clusters grows is due to Liang and Zeger (1986), and Exercise (C1) derives the same matrix as a bootstrap covariance. It recovers no efficiency and needs many independent units, as Section 31.4 shows; the two repairs are complementary.

31.2.6 Exercises

31.2.6.1 A. Check your understanding

Exercise. A1

Show directly from [Theorem 31.2.2\(a\)\(v\)](#) that ordinary least squares is best when the columns of $\mathbf{X}$ are eigenvectors of $\mathbf{V}$, and use this to recover the equicorrelation result of Exercise (A1).

Solution. If each column of $\mathbf{X}$ is an eigenvector of $\mathbf{V}$, then $\mathcal{C}(\mathbf{X})$ is spanned by eigenvectors, which is (v). For $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ the vector $\mathbf{1}$ is an eigenvector with eigenvalue $1 + (n - 1)\rho$ and every vector orthogonal to $\mathbf{1}$ is an eigenvector with eigenvalue $1 - \rho$. A model containing an intercept has $\mathcal{C}(\mathbf{X}) = \text{span}\{\mathbf{1}\} \oplus (\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp)$, a span of eigenvectors, so (v) holds.

Exercise. A2

Show that $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{M}$ for every $\mathbf{X}$ if and only if $\mathbf{V}$ is a multiple of $\mathbf{I}$.

31.2.6.2 B. Practice

Exercise. B1

Assume the simple covariance structure $\mathbf{V} = \mathbf{X}\mathbf{A}\mathbf{X}^\top + \mathbf{W}\mathbf{B}\mathbf{W}^\top$ of [Theorem 31.2.2\(a\)\(iv\)](#). Show that $\text{Cov}(\mathbf{M}\mathbf{Y}, (\mathbf{I} - \mathbf{M})\mathbf{Y}) = \mathbf{0}$, and that the error can be written $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_1 + \boldsymbol{\varepsilon}_2$ with $\boldsymbol{\varepsilon}_1 \in \mathcal{C}(\mathbf{X})$, $\boldsymbol{\varepsilon}_2 \in \mathcal{C}(\mathbf{X})^\perp$ and $\text{Cov}(\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2) = \mathbf{0}$ with probability one.

Solution. $\text{Cov}(\mathbf{M}\mathbf{Y}, (\mathbf{I} - \mathbf{M})\mathbf{Y}) = \sigma^2\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ by (ii). Put $\boldsymbol{\varepsilon}_1 = \mathbf{M}\boldsymbol{\varepsilon}$ and $\boldsymbol{\varepsilon}_2 = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$; these lie in the stated spaces, add to $\boldsymbol{\varepsilon}$, and have cross-covariance $\sigma^2\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. Their covariance matrices are $\sigma^2\mathbf{X}\mathbf{A}\mathbf{X}^\top$ and $\sigma^2\mathbf{W}\mathbf{B}\mathbf{W}^\top$.

Exercise. B2

Suppose $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$ with $\boldsymbol{\theta}$ unknown, and that $\mathcal{C}(\boldsymbol{\Sigma}(\boldsymbol{\theta})\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ for every $\boldsymbol{\theta}$ in the parameter set. Show that $\mathbf{M}\mathbf{Y}$ is a BLUE for every $\boldsymbol{\theta}$, so that the best estimator is known even though the covariance is not. Give an example with two unknown variance components.

Exercise. B3

In [Example \(Unequal clusters\)](#) with two clusters of sizes m_1 and m_2 , show that the efficiency of the sample mean as an estimate of the level is

$$e = \frac{(m_1 + m_2)^2}{\{(1 - \rho)(m_1 + m_2) + \rho(m_1^2 + m_2^2)\}(w_1 + w_2)}, \quad w_k = \frac{m_k}{1 - \rho + m_k\rho}.$$

Show that $e = 1$ when $m_1 = m_2$, and find the limit as $\rho \rightarrow 1$ with $m_1 = 1$ and $m_2 = m$.

Solution. $\text{Var}(\bar{Y}) = \sigma^2 \mathbf{1}^\top \mathbf{V} \mathbf{1} / n^2$ with $\mathbf{1}^\top \mathbf{V} \mathbf{1} = (1 - \rho)n + \rho \sum_k m_k^2$, and $\text{Var}(\hat{\mu}_{\mathbf{V}}) = \sigma^2 / \sum_k w_k$ by Eq. 31.1.8; divide. If $m_1 = m_2 = m$ both variances are $\sigma^2(1 - \rho + m\rho)/(2m)$, so $e = 1$. With $m_1 = 1$, $m_2 = m$ and $\rho \rightarrow 1$, $w_k \rightarrow 1$ and $e \rightarrow (1 + m)^2 / \{2(1 + m^2)\}$, which falls to $1/2$ as m grows.

31.2.6.3 C. Going deeper

Exercise. C1

Let $\text{rank}(\mathbf{X}) = p$ and put $\mathcal{B} = \{\boldsymbol{\lambda} \in \mathbb{R}^p : \mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X})\}$, the set of coefficient functions that ordinary least squares estimates best. Show that $\mathcal{B}$ is a subspace, that $\mathcal{B} = \mathbb{R}^p$ iff $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, and that $\dim \mathcal{B} = p - \text{rank}((\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1})$. Evaluate it for [Example \(Unequal clusters\)](#).

Solution. A vector $\boldsymbol{\lambda}$ lies in $\mathcal{B}$ iff $(\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} = \mathbf{0}$, so $\mathcal{B}$ is the null space of that matrix, hence a subspace, and Theorem 1.2.2 gives its dimension. It is all of $\mathbb{R}^p$ iff the matrix is zero, which is $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ because $(\mathbf{X}^\top \mathbf{X})^{-1}$ is nonsingular. In [Example \(Unequal clusters\)](#), $p = 2$ and the map kills the slope direction but not the level direction, so $\dim \mathcal{B} = 1$: the functions estimated best are exactly the multiples of the slope.

Exercise. C2

Suppose $\mathbf{V}$ has only two distinct eigenvalues, λ_1 with multiplicity k and λ_2 with multiplicity $n - k$, and let $\mathcal{E}_1$ be the eigenspace of λ_1 . Show that $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{M}$ iff $\mathcal{C}(\mathbf{X}) = (\mathcal{C}(\mathbf{X}) \cap \mathcal{E}_1) \oplus (\mathcal{C}(\mathbf{X}) \cap \mathcal{E}_1^\perp)$. Deduce that when this splitting holds every $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ has efficiency 1, and that when it fails there is a design attaining the bound $4\lambda_1\lambda_2/(\lambda_1 + \lambda_2)^2$.